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```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from sklearn.preprocessing import StandardScaler

import matplotlib.pyplot as plt
from sklearn import tree
```

```
df = pd.read_csv('/content/Admission_Predict_Ver1.1.csv')
print(df.head())
```

	Serial No.	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	\
0	1	337	118	4	4.5	4.5	9.65	
1	2	324	107	4	4.0	4.5	8.87	
2	3	316	104	3	3.0	3.5	8.00	
3	4	322	110	3	3.5	2.5	8.67	
4	5	314	103	2	2.0	3.0	8.21	

	Research	Chance of Admit
0	1	0.92
1	1	0.76
2	1	0.72
3	1	0.80
4	0	0.65

▼ =====

3. DATA PREPROCESSING

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```
df.drop(['Serial No.'], axis=1, inplace=True)

df['Admitted'] = df['Chance of Admit '] >= 0.75
df['Admitted'] = df['Admitted'].astype(int)

df.drop(['Chance of Admit '], axis=1, inplace=True)
```

```
print("\nProcessed Data:")
print(df.head())
```

Processed Data:

	GRE Score	TOEFL Score	University Rating	SOP	LOR	CGPA	Research	\
0	337	118	4	4.5	4.5	9.65	1	
1	324	107	4	4.0	4.5	8.87	1	
2	316	104	3	3.0	3.5	8.00	1	
3	322	110	3	3.5	2.5	8.67	1	
4	314	103	2	2.0	3.0	8.21	0	

Admitted

0	1
1	1
2	0
3	1
4	0

=====

4. FEATURE SELECTION

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```
X = df[['GRE Score', 'CGPA']]
y = df['Admitted']
```

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5. TRAIN-TEST SPLIT

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```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

6. FEATURE SCALING

```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
# =====
# 7. TRAIN MODEL (Decision Tree)
# =====
```

```
model = DecisionTreeClassifier(
    criterion='gini',
    max_depth=4,
    random_state=42
)
```

```
model.fit(X_train, y_train)
```

▼ DecisionTreeClassifier ⓘ ?
DecisionTreeClassifier(max_depth=4, random_state=42)

8. PREDICTION

```
y_pred = model.predict(X_test)
```

9. EVALUATION

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```

print("\nAccuracy:", accuracy_score(y_test, y_pred))

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

```

Accuracy: 0.88

Classification Report:

	precision	recall	f1-score	support
0	0.92	0.89	0.90	64
1	0.82	0.86	0.84	36
accuracy			0.88	100
macro avg	0.87	0.88	0.87	100
weighted avg	0.88	0.88	0.88	100

Confusion Matrix:

```

[[57  7]
 [ 5 31]]

```

▼

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10. VISUALIZE DECISION TREE

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```

plt.figure(figsize=(12,8))
tree.plot_tree(
    model,
    feature_names=['GRE Score', 'CGPA'],
    class_names=['Not Admitted', 'Admitted'],
    filled=True
)
plt.show()

```

